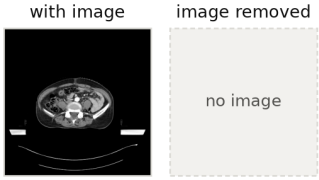
Consistent but Blind? Image Removal Misranks Image Reliance in Medical Vision-Language Models

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Code and data: https://github.com/thaitri2005/Medvlm-Image-Reliance

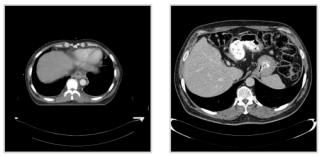
(a) "Is there spinal cord in this picture?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	quadrant (removed)
MedGemma-4B	yes	yes, yes	yes	Dangerous
Gemma-3-4B	yes	yes, yes	"Please provide the picture."	Ideal

Gemma-3 asks for the image; its forced yes/no choice is "no", so the probe counts it as image-reliant.

(b) "Does the picture contain heart?" (SLAKE CT, gold: yes)



	with image	2 paraphrases	image removed	wrong image	quadrant (removed)	quadrant (wrong)
MedGemma-4B	yes	yes, yes	yes	no	Dangerous	Ideal
Gemma-3-4B	yes	yes, yes	no*	yes	Ideal	Dangerous

** only 51% of first-token probability on yes/no. Same question, opposite verdicts from the two probes.*

Figure 1. **Two real questions, two models, two probes.** (a) Without the image, medically tuned MedGemma-4B still answers, and the consistent-but-dangerous taxonomy calls the answer DANGEROUS; its general-purpose twin Gemma-3-4B asks for the image, and its forced yes/no choice is scored as image-reliant (IDEAL). (b) On the same question, removing the image and showing a wrong image (whose correct answer is *no*) give opposite verdicts for both models; MedGemma follows the image when one is present. Real model outputs; SLAKE CT images (CC-BY-4.0) [13].

Abstract

The consistent-but-dangerous taxonomy flags a medical vision-language model (VLM) answer as DANGEROUS *when it is stable under paraphrase and unchanged when the image is removed. We ask whether this image-removal test can rank models by how much they use the image. We pose 3,365 questions over 907 CT, MRI, X-ray and pathology images to three medical VLMs and their general-purpose bases (MedGemma-4B and MedGemma-1.5-4B* vs. *Gemma-3-4B; MedVLThinker-3B* vs. *Qwen2.5-VL-3B) under five probes, with pre-registered endpoints, and with the probes and the replication pairs registered before they were run. For MedGemma-4B, image removal ranks the medical model as the less image-reliant one (*DANGEROUS *share +13.7 points [10.0, 17.5]), while every probe that keeps an image in the input ranks it as more reliant: it changes its answer more often when the image is swapped (+5 points), blacked out (+20), or replaced by one on which the same question has the opposite answer (73%* vs. *25%; on SLAKE, which both MedGemma releases were trained on, so its replication outside their training data is registered). The reversal replicates for MedGemma-1.5 (+37 points on opposite-answer images, −7 under image removal). It traces to refusal: forced to answer yes or no without an image, Gemma-3 would rather ask for the im-*

age on 29% of questions, and its forced answers there are unrelated to its answers with the image. In the Qwen pair neither model refuses, the in-format probes find no difference, and there is no reversal. Image removal also confounds image neglect with priors that agree with the image: MedGemma's excess DANGEROUS *answers are all correct. The* DANGEROUS *share rests on a chance floor of 36–42% on yes/no questions, and image reliance does not flag wrong answers (AUROC 0.51). Comparisons of image reliance between models need in-format probes.*

1. Introduction

A model that answers a question about an image without using the image is not doing visual question answering, however accurate it is. In general-domain VQA this concern produced balanced benchmarks and prior-robust splits [1, 8]. Medical VQA is still mostly evaluated by accuracy, where it matters more: a correct answer reached from language priors will be wrong on exactly the patients whose images are unusual.

Accuracy cannot separate *seeing* from *guessing well*, and consistency under rephrasing cannot either: a model that ignores the image is often *more* consistent, because nothing visual perturbs it [20]. Sadanandan and Behzadan [19] therefore crossed the two per sample. *Consistent* means the answer is stable under paraphrase; *image-reliant* means it changes when the image is removed. This gives four quadrants: IDEAL (consistent, reliant), FRAGILE, DANGEROUS (consistent but unchanged without the image) and WORST. On chest X-rays they found that fine-tuning for consistency moves most answers into DANGEROUS.

A taxonomy like this invites comparisons between models, and a comparison is only as good as the probe behind it. Removing the image changes the input format: a model asked about an image that is not there may refuse, answer from priors, or guess, and none of these is the same as ignoring an image that is present. We therefore ask whether image removal ranks two models the same way as probes that keep an image in the input. We audit MedGemma-4B [21], a small open medical VLM, against Gemma-3-4B [7], which has the same architecture and size but a general-purpose post-training, on CT, MRI, X-ray and pathology questions from three public benchmarks, and replicate on two further pairs: MedGemma-1.5-4B against the same base, and MedVLThinker-3B, a medical fine-tune of Qwen2.5-VL-3B, against its base. The design, endpoints and exclusions were frozen before the main run; three further probes, and then the replication pairs, were registered before they were run.

Figure 2 summarizes the answer. Image removal says MedGemma uses the image less than Gemma-3. A wrong image, a black image and an image on which the same question has the opposite answer all say it uses the image more. We trace the disagreement to two properties of image removal: it turns refusals into scored answers, and it cannot distinguish a model that ignores the image from one whose prior agrees with it. The reversal replicates for MedGemma-1.5 and is absent in the Qwen pair, the one pair whose models never refuse (Fig. 8).

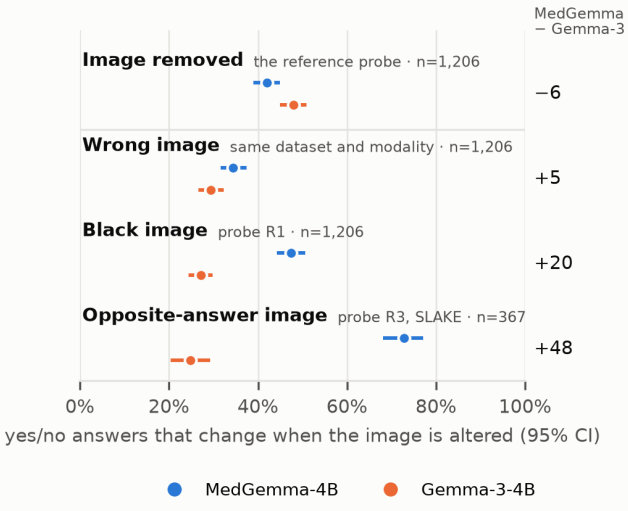


Figure 2. **The probes rank the two models in opposite directions.** Share of yes/no answers that change when the image is altered, by probe (95% CIs clustered by image; right: MedGemma − Gemma-3 in points). A model that uses the image should change its answer when the image changes. Only image removal, the probe that defines the taxonomy, ranks Gemma-3 as the more image-reliant model.

Contributions.

- A side-by-side comparison of four reliance probes on the same questions for three medical/general model pairs, showing that image removal ranks two pairs opposite to three in-format probes, with a pre-registered replication. Each probe has precedent [15, 19, 23]; to our knowledge their rankings of models have not been compared.
- Two confounds that explain the reversal, each with direct evidence: refusal under a forced yes/no format, which separates the pairs that reverse from the one that does not, and agreement between prior and image.
- Two corrections to the DANGEROUS share: its chance floor on yes/no questions (with Cohen's κ), and a paired split into correct and wrong DANGEROUS answers.
- A pre-registered, open pipeline across three datasets, four modalities and five models, with a paraphrase set screened pair by pair for equivalence and a finding that bf16 batching perturbs log-probabilities by up to 1 nat.

2. Related Work

Language priors in VQA. Balanced pairs [8] and changing-prior splits [1] showed that general VQA models exploit answer priors; text-only baselines are the medical analogue.

Does a medical VLM use the image? PSF-Med [20] measured paraphrase flip rates across medical VLMs on chest X-rays and noted that low flip rates often come with high text-only agreement. The quadrant taxonomy [19] defines reliance by image removal and corroborates it with a KL-divergence score, a black image and random image swaps; on their models these checks agreed with image removal. Zafar *et al*. [23] build blank, pixel-shuffled, absent and CLIP-retrieved hard-negative images to define reliance metrics and train a model against them. Lotfinia *et al*. [15] swap and occlude chest radiographs and find that model class (medical or general) does not predict image use; a question-only medical model scores close to multimodal ones. Pathology VLMs reach high accuracy without the image [25], and attention-based studies find that medical MLLMs often attend to irrelevant regions [14]. Grounding inputs can even hurt closed-ended accuracy [3]; HEAL-MedVQA [16] and MedHEval [2] benchmark localization and hallucination. These works use one probe or treat several as interchangeable; we compare the conclusions they reach about the same pair of models.

Answering without the evidence. MedVIGIL [11] tests whether models refuse when visual evidence is broken, and Vohra [22] finds that frontier VLMs confabulate diagnoses for images that were never attached. Four of our five small open models decline almost always when allowed to; we show what forcing an answer does to a reliance probe.

Selective prediction. Ranking answers by a score and abstaining on the lowest is evaluated by AUROC and risk–coverage [6]; we use it to ask whether reliance flags errors.

3. Method

3.1. Data

SLAKE [13] (English; CT, MRI, X-ray; 2,114 questions from test and validation, the only dataset with modality labels; validation is added so each modality has 53–82 images). **VQA-RAD** [12] (radiology, clinician-written; all 451 test questions, from the original release). **PathVQA** [9] (pathology, generated from textbook captions; 600 yes/no and 200 open test questions sampled with a fixed seed). A question is yes/no if its gold answer is *yes* or *no*. The **primary analysis set** is the 1,206 yes/no *test* questions on 670 images. Sampling, partner images and image hashes are fixed in committed manifests (SHA-1 seeded), so every run sees identical inputs.

3.2. Models

MedGemma-4B (`google/medgemma-4b-it`) is an open medical VLM whose SigLIP-style encoder [24] and language model were further trained on medical data; its technical report [21] lists SLAKE (450 items, the size of SLAKE's train split) and VQA-RAD (reinforcement-learning stage) among its training data and states that PathVQA was removed. **Gemma-3-4B** (`google/gemma-3-4b-it`) shares its architecture and size. The two differ in their whole post-training, not only in medical data, so we compare *checkpoints* and do not attribute differences to medical data alone. **Replication pairs**, registered before any of their outputs were seen: **MedGemma-1.5-4B** (`google/medgemma-1.5-4b-it`, the 2026 release with the same architecture; its card lists SLAKE among training data) against the same Gemma-3-4B, and **MedVLThinker-3B** (`UCSC-VLAA/MedVLThinker-3B-RL_PMC`, Qwen2.5-VL-3B-Instruct trained with reinforcement learning on PMC-VQA) against its base **Qwen2.5-VL-3B-Instruct**. All five models run unquantized in bf16 on one 16 GB GPU (quantization would perturb the probabilities we audit) and receive identical prompts: a fixed system line, the question, and "Answer with only 'yes' or 'no'." or "Answer with a single word or short phrase." Each new model passed a registered format check before its main run: mean yes/no first-token mass with the image ≥ 0.9 in every dataset of the 330-question pilot (0.93–0.999).

3.3. Probes

Every question is posed with its image (**orig**) and with screened paraphrases (**para**$_k$, original image; Sec. 3.5). Reliance is probed five ways, each differing from *orig* only in the image or the instruction:

- **Image removed** (*noimg*): image tokens omitted, text identical; the reference probe of the taxonomy [19].
- **Wrong image** (*shuf*): another image from the same dataset and modality, chosen by a hash of the question ID.
- **R1, black image**: a black image in the image slot, in format but without content.
- **R2, "unknown" allowed**: image removed, and the instruction adds "If this cannot be determined without seeing the image, answer 'unknown'."
- **R3, opposite-answer image**: another image of the same dataset and modality on which the *same* question has the opposite gold answer. A model that uses the image must change its answer. Partners exist for 367 SLAKE yes/no questions (130 images), whose templated questions recur

across images.

The first two belong to the pre-registered main run. R1–R3 were registered after it, with directional hypotheses (H6: MedGemma's answers survive a black image more; H7: with "unknown" allowed MedGemma still declines less; H8: MedGemma follows opposite-answer images more), and run afterwards.

3.4. Measures

Yes/no answers come from one forward pass: $P(\text{yes}) = \sigma(\text{LSE}_{t\in Y}\,\ell_t - \text{LSE}_{t\in N}\,\ell_t)$ over the single-token variants of *yes* and *no*; the first-token mass $m = P(Y) + P(N)$ is logged as a format check. **Open answers** use greedy decoding and token-F1 ≥ 0.5 after normalization, with an opposite-term guard (left/right, upper/lower, *etc*. never match). **Consistent**: the answer on the first two valid paraphrases equals the *orig* answer. **Reliant** under a probe: the probe answer differs from *orig*. The **quadrant** follows from (consistent, reliant). **Refusal**: $m < 0.5$, *i.e*. most first-token mass is on neither *yes* nor *no*; free generation confirms these are requests for the image (Fig. 1a). **Image PMI** of an answer $\hat{y}$ is $\log p(\hat{y} \mid I, q) - \log p(\hat{y} \mid \varnothing, q)$. **Chance correction**: Cohen's κ [4] between *orig* and probe answers; the *chance floor* of the DANGEROUS share is $P(\text{consistent}) \cdot p_e$, where p_e is the agreement expected from the two answer distributions alone.

3.5. Paraphrase construction and screening

Paraphrase consistency is meaningful only if a paraphrase cannot change the correct answer. Candidates come from Qwen2.5-3B-Instruct [18] (greedy; five styles) and from two deterministic templates that copy every content word ("Is there X?"→"Is X present?"). A rule validator rejects changes to laterality, negation and absence words, quantities, antonyms, imaging planes, question form, answer leakage, near-copies and length. Two audit rounds accepted 81% and 83%, below the pre-specified 85%, so we tightened the rules, excluded a "plain language" style, and **judged every unique rule-valid (question, paraphrase) pair**, 4,890 in total, against a written rubric (Appendix B). The judge was an LLM (Claude); see Sec. 6. 1,208 pairs were rejected, each with a recorded reason (*e.g*. *mass→lesion*, *is enlarged→can be enlarged*), and unjudged pairs are rejected by construction. 84% of the primary-set questions retain at least two valid paraphrases; quadrant analyses use those (1,012 questions, 600 images).

3.6. Statistics and pre-registration

Hypotheses, endpoints, tests and exclusions were frozen before the main run [17]; every later change is listed in Appendix C, and post-hoc analyses are labeled as such. Several questions share an image, so all intervals are percentile 95% CIs from a cluster bootstrap over images [5] ($B = 2000$) with two-sided bootstrap p-values. Endpoints (Tab. 2): **P1a–c** blind accuracy minus the majority-answer prior per dataset and **P2** AUROC(image PMI) − AUROC(confidence), one Holm family [10]; **P3** the paired difference in DANGEROUS share, MedGemma − Gemma-3; and **H6–H8**, a second Holm family; for the replication pairs (medical − general), **H9** follows the opposite-answer image more (> 0), **H10** changes its answer less without the image (< 0) and **H11** refuses less without the image (< 0), a third Holm family. A pair replicates the reversal if H9 and H10 hold. A finding is called robust only if its sign holds across a grid of F1 thresholds (0.3/0.5/0.8), paraphrase counts (1/2/3) and "strong" flips.

4. Results

Both runs completed (18,896 calls per model, plus 3,535 per model for R1–R3). Our pipeline reproduces MedGemma's published SLAKE score (token-F1 70.2 [67.5, 72.7] *vs*. 72.3 on the model card; Gemma-3: 45.1 *vs*. 40.2), and the yes/no format mass with the image is ≥ 0.997 in every dataset. MedGemma is the more accurate model with the image (Tab. 1), by 18 points on SLAKE and 25 on VQA-RAD yes/no.

4.1. The reference probe: MedGemma looks more consistent but blind

Under image removal MedGemma's DANGEROUS share is 50.6/46.8/56.5% (PathVQA/SLAKE/VQA-RAD) against 43.2/28.1/31.0% for Gemma-3 (Fig. 3, left). The paired difference, **P3** $= +13.7$ **points** [10.0, 17.5] ($p < 0.001$; 1,012 questions, 600 images), is positive in every dataset (PathVQA +7.5 [2.6, 12.3], SLAKE +18.6 [12.2, 24.9], VQA-RAD +25.6 [14.7, 36.5]) and in every cell of the sensitivity grid. MedGemma is also the more consistent model (paraphrase flip rate 6.8% *vs*. 13.6% on SLAKE). Read at face value, the medical checkpoint is more often stable, confident and blind.

4.2. In-format probes reverse the ranking

Figure 2 places all four probes on one scale, the share of answers that change. Image removal is the only one under which Gemma-3 changes more (-6.1 [-9.5, -2.7] points). The three in-format probes all say the opposite.

Opposite-answer image (R3). This is the most direct test: the question is unchanged and the image now calls for the other answer. MedGemma follows the image on 72.8% [68.1, 77.0] of swaps and Gemma-3 on 24.8% [20.4, 29.3] (H8, +48.0 points), so MedGemma is right on the new image 86% of the time and Gemma-3 61%. The gap holds for CT (75 *vs*. 26%), MRI (57 *vs*. 26%) and X-ray (81 *vs*. 8%; Fig. 4) and when 40 PathVQA and VQA-

Table 1. Accuracy (%) on test questions. Yes/no: with the image, without it ("none"), with a wrong image, against the majority-answer prior. Open: token-F1 ≥ 0.5 with the image. CIs are in the repository.

		MedGemma-4B (yes/no)			Gemma-3-4B (yes/no)			Open, with image	
Dataset	Prior	image	none	wrong	image	none	wrong	MedGemma	Gemma-3
PathVQA (pathology)	52.2	65.5	57.2	51.0	63.2	53.5	46.7	10.5	4.0
SLAKE (CT/MRI/X-ray)	50.7	**86.5**	47.6	67.0	68.7	47.6	62.3	66.7	37.1
VQA-RAD (radiology)	53.0	**81.3**	60.6	70.5	56.2	55.4	60.2	46.0	33.5

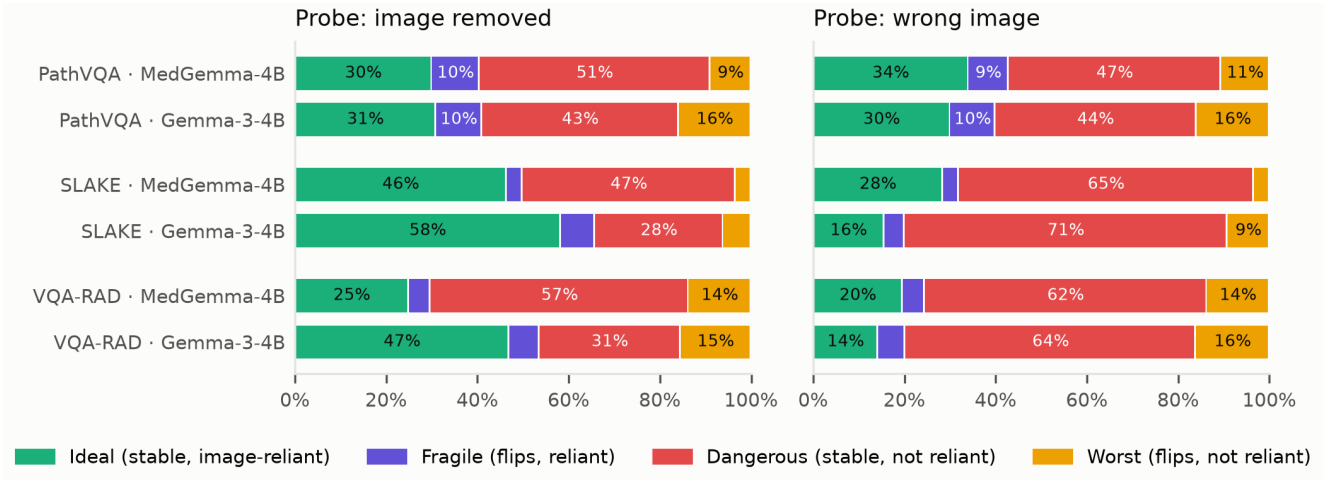


Figure 3. Quadrant composition of yes/no answers (test). Left: reliance probed by removing the image (the reference definition). Right: by a wrong image of the same modality. Shares are raw; on yes/no questions they include chance agreement (Sec. 4.4) and correct answers (Sec. 4.3).

Table 2. Confirmatory tests. P1–P2 concern MedGemma (P2 on the model's own answer, Appendix C); P3 and H6–H8 are paired differences, MedGemma − Gemma-3 (points, or κ for H6). 95% CIs clustered by image.

	Test	Estimate	Outcome
Frozen before the main run			
P1a	blind − prior, PathVQA	$+5.0\,[0.9, 8.8]$	p_{Holm}=.042
P1b	SLAKE	$-3.1\,[-7.8, 1.8]$	n.s.
P1c	VQA-RAD	$+7.6\,[0.4, 14.5]$	p_{Holm}=.072
P2	AUROC: PMI − conf.	$-0.20\,[-0.25, -0.15]$	worse
P3	DANGEROUS, image removed	$+13.7\,[10.0, 17.5]$	$p<.001$
Registered after the main run, before the probes			
H6	κ(image, black) > 0	$-0.22\,[-0.30, -0.14]$	reversed
H7	declines, "unknown" < 0	$-6.7\,[-8.2, -5.3]$	supported
H8	follows opp. image > 0	$+48.0\,[41.9, 53.8]$	supported

RAD swaps are added (69.8 *vs.* 24.8%, exploratory). Following is reading, not flipping: the two gold answers are opposite, so a changed answer is right on both images or wrong on both. MedGemma is right on both (*pair accuracy*, 0 by construction for a model that ignores the image) on 71.7% [67.1, 75.8] of swaps and wrong on both on 1.1%; Gemma-3 gives the same answer to both images on 75.2%. Resampling SLAKE's 53 question templates instead of images widens the interval of the gap to [36.4, 57.7] (post hoc). SLAKE is, however, training data of both MedGemma releases (Sec. 6).

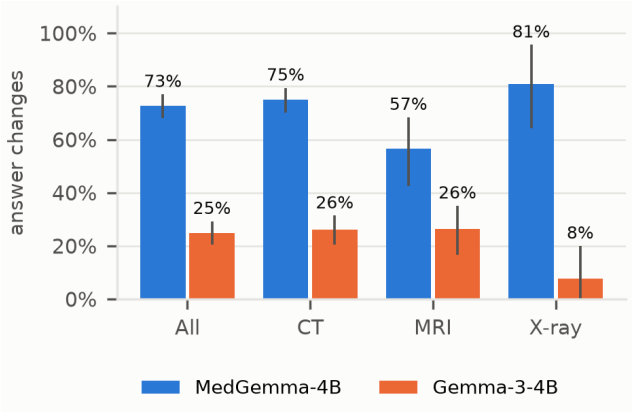


Figure 4. Probe R3: share of answers that change when the image is replaced by one on which the same question has the opposite correct answer (SLAKE yes/no, test+validation; 95% CIs by image).

Black image (R1). H6 predicted that MedGemma's answers would survive a black image more often; it failed in the opposite direction. Gemma-3 keeps the answer it gave with the real image on 72.9% of questions, MedGemma on 52.6%; chance-corrected, $\kappa = 0.33\,[0.28, 0.39]$ *vs.* $0.11\,[0.06, 0.17]$. Under this probe Gemma-3 is the more DANGEROUS model by 14.0 points [9.4, 18.6]. The answers show why: on a black image MedGemma's yes-rate drops from 58% to 28% (it reads an empty image as absent findings), while Gemma-3's rises from 64% to 84%, tracking the question rather than the image.

Wrong image. A random image of the same modality changes MedGemma's answer more often (+5.1 points [1.7, 8.5]); the DANGEROUS share under this probe does not differ ($-0.7\,[-4.4, 3.2]$). It is the weakest probe, because a random partner often has the same correct answer: where the partner image carries the same question, *all 23* chest X-ray partners had the same answer (CT 58%, MRI 60%). This explains why the modality ordering of the DANGEROUS share reverses between probes (SLAKE, MedGemma): X-ray is the least DANGEROUS modality under image removal (33.1%) and the most under a wrong image (75.4%). The two probes agree poorly (κ between their reliance labels -0.11 to 0.23).

4.3. Why image removal misranks

Refusal. The pre-registered format check ($m \geq 0.9$) failed in exactly one condition, image removed. Free generation explains it: "*Please provide the picture. I need to see the image to determine if there is a spinal cord present*" (Gemma-3). Gemma-3 would rather refuse on 354 of the 1,206 questions and MedGemma on 219 (Fig. 5; 14/41/50% *vs.* 10/26/27% by dataset). On those questions the forced answer is mostly *no* and unrelated to the model's answer with the image ($\kappa = -0.01$ for Gemma-3, 0.04 for MedGemma), so 53–55% of them count as "image-reliant". Allowed to say "unknown" (R2), MedGemma declines on 93.0% and Gemma-3 on 99.7%. H7's direction holds (-6.7 points) and MedGemma keeps a small tendency to answer blind, concentrated in PathVQA (12% answered), but most of the forced-format difference of 11.2 points disappears.

Agreement between prior and image. Refusal is not the whole story: on the 436 questions where both models answer without the image ($m \geq 0.9$), P3 is still +8.0 [2.9, 13.3]. Here MedGemma's blind answers are informative. It beats the majority-answer prior on PathVQA (P1a, +5.0; +11.9 [7.1, 16.6] among questions it answers), and its no-image answers agree with its image answers beyond chance ($\kappa = 0.21$ *vs.* 0.11 for Gemma-3, each on the questions it answers). A model that looks at the image and whose prior predicts what the image shows

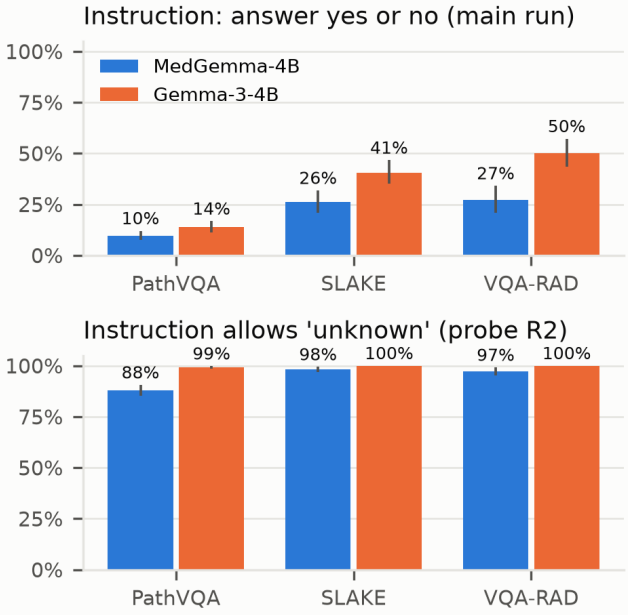


Figure 5. Share of yes/no questions on which the model declines to answer when the image is removed ($m < 0.5$; 95% CIs by image). Top: the main run's instruction ("Answer with only yes or no"). Bottom: probe R2, whose instruction allows "unknown".

gives the same answer with and without it, and image removal scores that as blind. The correctness split shows that this is what MedGemma's excess consists of (Fig. 6, bottom): DANGEROUS-and-correct is 39.2% *vs.* 25.1% (+14.1 [11.1, 17.1]), while DANGEROUS-and-wrong, the cell that would mislead a user, is 11.3% *vs.* 11.7% ($-0.4\,[-2.7, 2.0]$). R3 removes this case by construction, and there MedGemma follows the image. Only the PathVQA margin is free of training exposure: PathVQA was removed from MedGemma's training, whereas VQA-RAD was used, and 202 of our 203 VQA-RAD test images also occur in VQA-RAD's train split.

4.4. The chance floor

An answer counts as image-independent whenever the no-image answer equals the with-image answer, which on yes/no questions happens often by chance. On the P3 set MedGemma's answers are unchanged without the image 58.6% of the time against 49.1% expected from its answer distributions alone (Gemma-3: 49.7% *vs.* 45.7%). Combined with consistency this gives a **chance floor of 41.5% (MedGemma) and 35.8% (Gemma-3)**, against observed DANGEROUS shares of 50.5% and 36.8%. About 6 of the 13.7 points of P3 are the difference in floors, partly because MedGemma is more consistent (84.5% *vs.* 78.4%), a desirable property the metric counts against it. Corrected for chance the image-removal effect remains, $\Delta\kappa = +0.112\,[0.044, 0.180]$, but it is the size of a consistency dif-

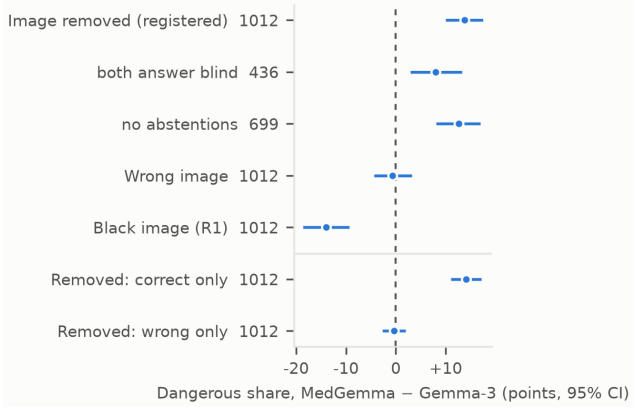


Figure 6. The MedGemma − Gemma-3 difference in DANGEROUS share under each probe and subset, and split by whether the DANGEROUS answer is correct (paired; 95% CIs by image).

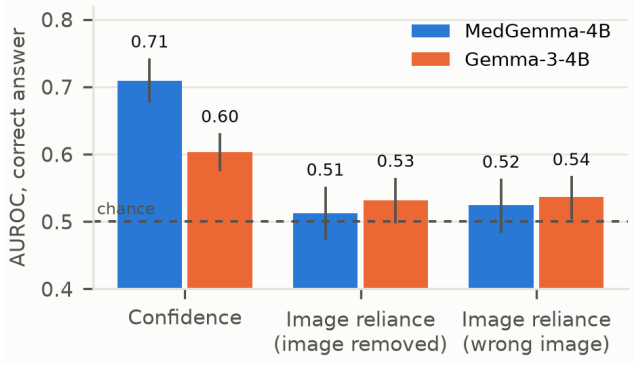


Figure 7. AUROC for ranking correct above incorrect yes/no answers (test, 95% CIs): the model's confidence *vs*. label-free image reliance under each probe.

ference, not of a model that ignores images.

4.5. Does image reliance flag wrong answers?

If image-independent answers were the unreliable ones, image reliance could serve as a warning signal. Ranking each yes/no answer by its image PMI separates correct from incorrect answers at chance (Fig. 7; AUROC **0.513** [0.473, 0.552] for MedGemma, 0.532 for Gemma-3), while the model's own confidence reaches 0.709 and 0.603 (P2: better by 0.197 [0.145, 0.250] and 0.071 [0.026, 0.116]). The wrong-image variant is no better (0.524, 0.536). Whether a model leaned on the image does not tell us whether it was right.

4.6. Replication on two more pairs

MedGemma-1.5 replicates the reversal. H9 and H10 hold ($p_{\text{Holm}} < 0.001$, Tab. 3). MedGemma-1.5 follows opposite-answer images on 61.3% of swaps against Gemma-3's 24.8% (+36.5 [30.6, 42.1]) and changes more under a black image (+8.2 [3.7, 12.6]), but changes *less* when the image is removed (−7.4 [−11.2, −3.7]), and its DANGEROUS share under image removal is 9.8 points [5.8, 13.9] higher. Refusal is even more lopsided than for MedGemma-4B: under the forced format it declines on 8 of 1,206 questions, Gemma-3 on 354 (H11, −28.7 [−31.6, −26.0]). On the 467 questions both models answer, the DANGEROUS gap shrinks to +3.0 [−3.6, 9.5], so refusal accounts for nearly all of it. Unlike MedGemma-4B, MedGemma-1.5 also answers blind when allowed to decline (it declines on only 52.7%), and its excess DANGEROUS answers are again correct (+9.3 [5.7, 12.6] correct, +0.5 [−2.1, 3.2] wrong).

Table 3. Replication tests, medical − general (points; 95% CIs in the text). Primary: MedGemma-4B *vs*. Gemma-3-4B; A: MedGemma-1.5-4B *vs*. Gemma-3-4B; B: MedVLThinker-3B *vs*. Qwen2.5-VL-3B. Forced-format refusal without the image, medical / general: 18.2 / 29.4%, 0.7 / 29.4%, 0 / 0%.

Pair	H9: opposite image	H10: removed	H11: refusal	Replicates
Primary	+48.0	−6.1	−11.2	(reference)
A	+36.5	−7.4	−28.7	yes
B	−1.6	+3.9	0.0	no

The Qwen pair does not. MedVLThinker and its base are indistinguishable under every in-format probe (opposite-answer −1.6 [−4.9, 1.4], black −0.8 [−3.0, 1.3], wrong −1.2 [−3.3, 1.0]) and in accuracy (error rate 30.8% *vs*. 31.7%; SLAKE yes/no 78.9% *vs*. 76.6%): the fine-tune did not change how the model uses the image. Neither model ever declines under the forced format, and both decline on at least 98.7% when allowed to. With no refusal difference there is nothing to reverse. Image removal still reports a small difference that the in-format probes do not see, in the opposite direction to H10 (+3.9 [1.7, 6.1]), because the base's blind answers agree more with its image answers ($\kappa = 0.16$ *vs*. 0.09). Across the three pairs, image removal matched the in-format probes in none; it reversed their ranking in the two pairs whose models differ in refusal.

5. Discussion

What image removal measures. Removing the image asks a model a question about an image that is not there. Its answer depends on whether it will answer at all and on how good its priors are, and only then on whether it would have used the image. In our pair these two factors point the same way: the general checkpoint refuses more and has weaker priors, so it looks more image-reliant, while every probe that keeps an image in the input finds the medical checkpoint more reliant, by up to 48 points. The original study's corroborating checks (black image, random swaps)

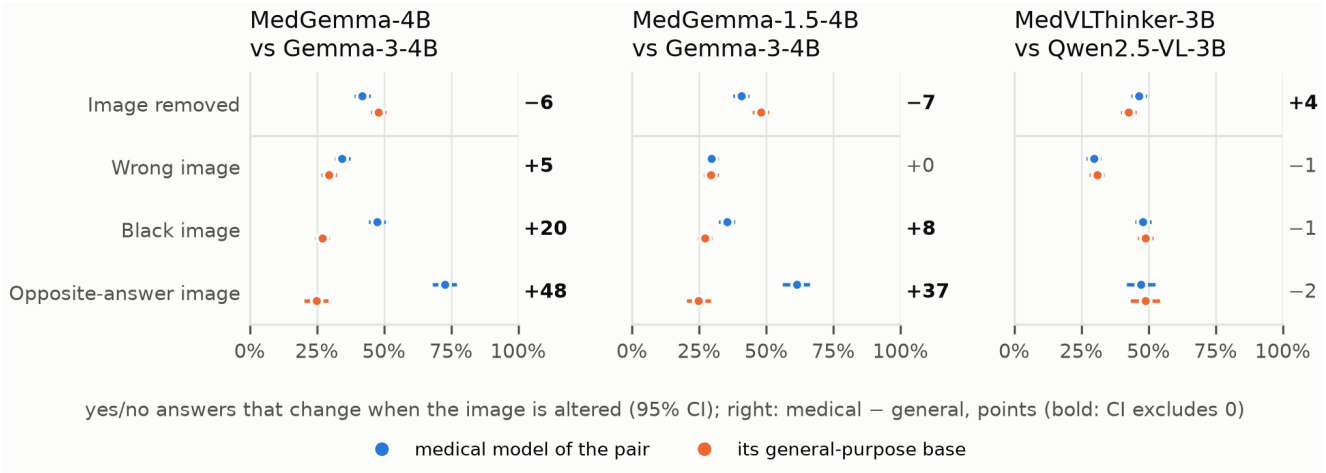


Figure 8. **The comparison of Fig. 2 for all three pairs.** Blue: the medical model; orange: its general-purpose base. Right of each panel: medical − general in points, bold where the 95% CI (clustered by image) excludes 0. Image removal ranks both MedGemma models below Gemma-3 while every in-format probe that separates the models ranks them above it; the Qwen pair differs under no in-format probe.

agreed with image removal on its chest X-ray models [19]; on our models and data they do not. Agreement between probes therefore has to be checked for each comparison rather than assumed, and a DANGEROUS share is underspecified without its probe. The replication pairs locate the failure: image removal reversed the ranking in both pairs whose models differ in how often they refuse, and not in the Qwen pair, whose models never refuse. Whether a medical checkpoint uses the image more than its base is itself checkpoint-specific: both MedGemma releases do, MedVLThinker does not.

Where the residual risk is. MedGemma's extra DANGEROUS answers are correct, and the stable, image-independent and wrong cell is the same for both models. The remaining concern is that MedGemma's errors, fewer overall (25.7% *vs*. 37.6%), are more often of this kind (44% *vs*. 31%, +12.9 [6.0, 19.9]), so a consistency or reliance check would miss them. It also answers without an image on 7% of questions even when allowed to decline.

Recommendations for reliance audits. (1) Probe reliance in format, ideally with an image that calls for a different answer, and report agreement between probes. (2) Report the DANGEROUS share with its chance floor or as κ; on balanced yes/no data a model whose blind answers are pure noise is about 50% "not reliant". (3) Split DANGEROUS by correctness. (4) Allow abstention, and treat declining without an image as desirable rather than as a format failure.

6. Limitations

Three pairs from two model families show that image removal *can* misrank models and that refusal predicts when; they do not estimate how often this happens across the field. The refusal account of which pairs reverse was formed after the primary results and fits all three pairs, but three pairs cannot test it formally. Medical and general checkpoints differ in their whole post-training, not only in medical data. The probes were registered after the main results were known (before running them), and one of three predictions failed. R3 is limited to SLAKE, where questions recur across images, and SLAKE is training data of both MedGemma releases but not of MedVLThinker: its evaluated images are not training images, but their distribution is, so a training-distribution effect explains R3 as well as greater use of the image does. A replication on 1,192 PathVQA questions, which no medical model lists as training data, is registered with its decision rule (preregistration §17) and has not been run at the time of writing. MedGemma-4B was also trained on VQA-RAD, whose test images almost all occur in its train split, so its VQA-RAD results may reflect exposure. Refusal is inferred from first-token mass and confirmed on a sample by free generation, not annotated per question. Paraphrase equivalence was judged by an LLM and **was not validated by human raters**; this is the main limitation of the consistency measure, and an LLM judge may share blind spots with the paraphraser. PathVQA questions are auto-generated and often ungrammatical. The correctness split, the chance correction and the refusal split were added after seeing the results. All 330 pilot questions used to tune the design are inside the evaluation set; excluding them leaves every conclusion unchanged (P1a +6.3 [2.4, 10.6], P3 +12.9 [8.9, 17.1]).

Benchmark VQA is not clinical use, and nothing here is a clinical claim.

7. Conclusion

Is a small medical VLM consistent but blind? By the image-removal test, MedGemma looks that way, more than its general-purpose twin. By every test that keeps an image in the input it is the opposite: it follows the image more, by up to 48 points, and so does MedGemma-1.5. Image removal measures whether a model answers without an image and whether its priors agree with the image; where two models differ in the first, it reverses their ranking. Comparisons of image reliance between models need in-format probes, chance correction and a correctness split.

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A. Implementation details

Pipeline. A Python package (`medvlm`) with a command-line interface: `prepare` (download, deterministic sampling, SHA-checked images), `paraphrase` (generation plus rule and judgment validation, fail-closed), `pipeline` (resumable inference in parts with checkpoint analysis), `probes` (R1–R3), `analyze`, `compare` and `posthoc` (all tables and figures here). Configs layer base, profile, model and hardware settings. Each job is one (question, condition) pair keyed by ID, so interrupted runs resume without recomputation, and every run records its config, commit and library versions.

Inference. PyTorch 2.11 and Hugging Face Transformers 5.19, bf16, SDPA attention, one RTX 4080 SUPER (16 GB). A full run is 18,896 calls per model plus 3,535 for the probes, for each of five models (MedGemma-4B: 0.23 s per call, about 75 minutes, 8.2 GB peak). During preflight we found that in bf16 the *batch size* changes teacher-forced log-probabilities by up to 1.09 nats (median 0.17) even with identical, unpadded inputs, while $P(\text{yes})$ changes by at most 0.008. An fp32 output head did not remove it, so the cause is inside the network. All runs therefore use batch size 1, which costs about 1.5× throughput. A regression check reproduced the T4 pilot on the new GPU (98.5% identical answers, mean $|\Delta P(\text{yes})| = 0.004$).

Testing. 97 unit tests and 7 real-model tests cover, among others, scoring edge cases, paraphrase rules (including each failure class found in the audit), batching equivalence, crash-safe resume, the coverage of the clustered bootstrap (by simulation), the test-split restriction of P3, chance agreement of κ, and the construction of opposite-answer partners.

B. Paraphrase equivalence rubric

A pair is **accepted** only if it asks exactly the same thing, so the correct answer cannot differ. Allowed: perception framing ("can X be seen", "is X visible", "can X be seen to be enlarged"), exact synonyms (clavicle/collarbone, encephalon/brain, intravenous contrast/dye), reordering. **Rejected**: a fact turned into a possibility or capability ("Is the heart enlarged?"→"Can the heart be enlarged?"); a broadened or narrowed term (mass→lesion, organs→structures, digestive system→GI tract, skull fracture→cranial injury); a changed scope or presupposition ("Do the organs exist in X"→"Are there organs in X"); added qualifiers, findings or modality ("bilaterally", "on imaging studies", "radiograph"); a reversed or changed relation ("does A show B"→"is A associated with B", "mass lateral to X"→"X lateral to mass"); antonyms (healthy↔unhealthy, coronal→transverse); two questions in one; malformed text. Rejections by class: relation 447, term 364, modal 106, added content 102, organ→structure 47, generic collapse 30, polarity 19, scope 9 (plus template-pass rejections).

C. Deviations from the pre-registration

Every change made after the analysis plan was frozen, with its reason.

Change	Reason and effect
Batch size 1 for all runs	bf16 batching shifts teacher-forced log-probabilities by up to 1.09 nats; decided before the main run.
Validator v2/v3, *plain* style excluded, full equivalence screen, template styles	Audit acceptance below the pre-specified 85% line; before the main run.
Format stop-the-line rule triggered (image removed only)	Model behavior (refusal), not a pipeline error; forced-choice $P(\text{yes})$ kept, refusal reported, P1/P3 re-estimated without refusals.
P2 computed on the model's answer $\hat{y}$	The registered formula used the gold answer, which is unavailable at deployment and predicts correctness by construction.
P3 restricted to the test split	An analysis bug had pooled the validation split; fixed to the registered set.
Chance correction, correctness split, refusal split, answer-change comparison across probes	Added after the run; descriptive, reported next to the registered estimates.
Robustness probes R1–R3	Registered after the main run, before running them (docs/PREREGISTRATION.md §15); R3 restricted to SLAKE as registered (the run also covered 40 other swaps, reported as exploratory).
Replication pairs, H9–H11	Registered after the primary results, before running (docs/PREREGISTRATION.md §16); all three new models passed the registered format check; the fallback pair was not needed.
Pair accuracy for R3	Post hoc: separates correct readings from answer changes; registered for R4.
Probe R4, H12a–c	Opposite-answer probe on PathVQA, outside the medical models' training data; registered before running (§17), pending.
Training-data statements	Corrected after re-checking the documentation (§18): MedGemma-4B was trained on VQA-RAD; MedGemma-1.5's card lists only SLAKE.